

1

Question 5

Consider $f(x) = x^2 + \frac{p}{x}$, $x \neq 0, p \in R$.

There is a stationary point on the graph of f when $x = -2$.

The value of p is

- A. -16
- B. -8
- C. 2
- D. 8
- E. 16

2

Question 10

For the curve $y = x^2 - 5$, the tangent to the curve will be parallel to the line connecting the positive x -intercept and the y -intercept when x is equal to

- A. $\sqrt{5}$
- B. 5
- C. -5
- D. $\frac{\sqrt{5}}{2}$
- E. $\frac{1}{\sqrt{5}}$

3

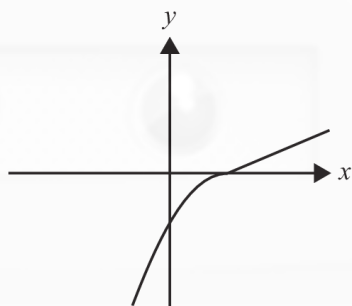
Consider the tangent to the graph of $y = x^2$ at the point $(2, 4)$.

Which of the following points lies on this tangent?

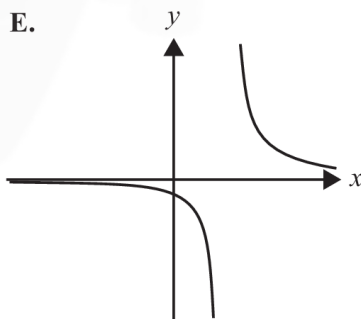
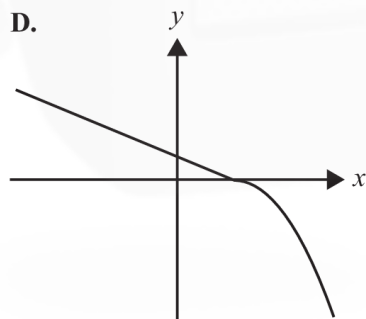
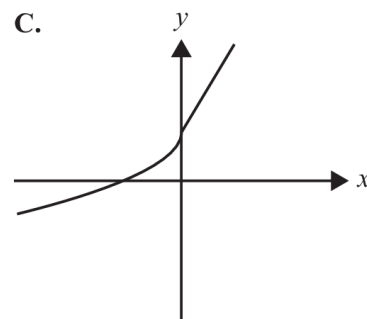
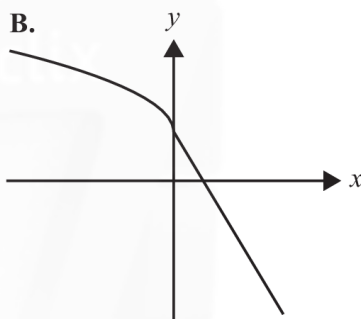
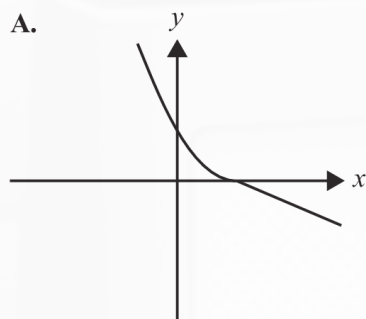
- A. $(1, -4)$
- B. $(3, 8)$
- C. $(-2, 6)$
- D. $(1, 8)$
- E. $(4, -4)$

Question 6

Part of the graph of the function f is shown below. The same scale has been used on both axes.



The corresponding part of the graph of the inverse function f^{-1} is best represented by



Question 5

Which one of the following is the inverse function of $g: [3, \infty) \rightarrow R$, $g(x) = \sqrt{2x-6}$?

- A. $g^{-1}: [3, \infty) \rightarrow R$, $g^{-1}(x) = \frac{x^2+6}{2}$
- B. $g^{-1}: [0, \infty) \rightarrow R$, $g^{-1}(x) = (2x-6)^2$
- C. $g^{-1}: [0, \infty) \rightarrow R$, $g^{-1}(x) = \sqrt{\frac{x}{2}+6}$
- D. $g^{-1}: [0, \infty) \rightarrow R$, $g^{-1}(x) = \frac{x^2+6}{2}$
- E. $g^{-1}: R \rightarrow R$, $g^{-1}(x) = \frac{x^2+6}{2}$

The function $f: D \rightarrow R$ with rule $f(x) = 2x^3 - 9x^2 - 168x$ will have an inverse function for

- A. $D = R$
- B. $D = (7, \infty)$
- C. $D = (-4, 8)$
- D. $D = (-\infty, 0)$
- E. $D = \left[-\frac{1}{2}, \infty\right)$

If $f(x-1) = x^2 - 2x + 3$, then $f(x)$ is equal to

- A. $x^2 - 2$
- B. $x^2 + 2$
- C. $x^2 - 2x + 2$
- D. $x^2 - 2x + 4$
- E. $x^2 - 4x + 6$

The graphs of $y = mx + c$ and $y = ax^2$ will have no points of intersection for all values of m , c and a such that

- A. $a > 0$ and $c > 0$
- B. $a > 0$ and $c < 0$
- C. $a > 0$ and $c > -\frac{m^2}{4a}$
- D. $a < 0$ and $c > -\frac{m^2}{4a}$
- E. $m > 0$ and $c > 0$

Question 20

A transformation $T: R^2 \rightarrow R^2$, $T\left(\begin{bmatrix} x \\ y \end{bmatrix}\right) = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} + \begin{bmatrix} 5 \\ 0 \end{bmatrix}$ maps the graph of a function f to the graph of $y = x^2$, $x \in R$.

The rule of f is

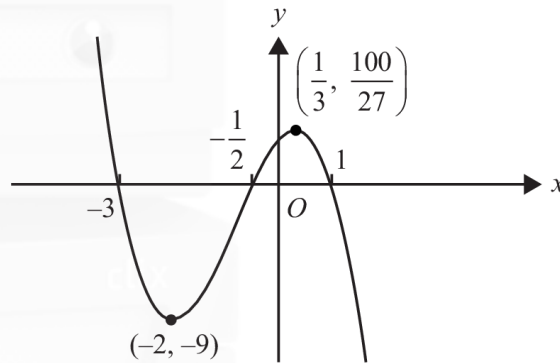
- A. $f(x) = -(x + 5)^2$
- B. $f(x) = (5 - x)^2$
- C. $f(x) = -(x - 5)^2$
- D. $f(x) = -x^2 + 5$
- E. $f(x) = x^2 - 5$

For the polynomial $P(x) = x^3 - ax^2 - 4x + 4$, $P(3) = 10$, the value of a is

- A. -3
- B. -1
- C. 1
- D. 3
- E. 10

Question 3

Part of the graph $y = f(x)$ of the polynomial function f is shown below.



$f'(x) < 0$ for

- A. $x \in (-2, 0) \cup \left(\frac{1}{3}, \infty\right)$
- B. $x \in \left(-9, \frac{100}{27}\right)$
- C. $x \in (-\infty, -2) \cup \left(\frac{1}{3}, \infty\right)$
- D. $x \in \left(-2, \frac{1}{3}\right)$
- E. $x \in (-\infty, -2] \cup (1, \infty)$

Let f be a function with domain R such that $f'(5) = 0$ and $f'(x) < 0$ when $x \neq 5$.

At $x = 5$, the graph of f has a

- A. local minimum.
- B. local maximum.
- C. gradient of 5.
- D. gradient of -5 .
- E. stationary point of inflection.

The transformation that maps the graph of $y = \sqrt{8x^3 + 1}$ onto the graph of $y = \sqrt{x^3 + 1}$ is a

- A. dilation by a factor of 2 from the y -axis.
- B. dilation by a factor of 2 from the x -axis.
- C. dilation by a factor of $\frac{1}{2}$ from the x -axis.
- D. dilation by a factor of 8 from the y -axis.
- E. dilation by a factor of $\frac{1}{2}$ from the y -axis.

Question 12

The graph of a function f is obtained from the graph of the function g with rule $g(x) = \sqrt{2x-5}$ by a reflection in the x -axis followed by a dilation from the y -axis by a factor of $\frac{1}{2}$. Which one of the following is the rule for the function f ?

- A. $f(x) = \sqrt{5-4x}$
- B. $f(x) = -\sqrt{x-5}$
- C. $f(x) = \sqrt{x+5}$
- D. $f(x) = -\sqrt{4x-5}$
- E. $f(x) = -\sqrt{4x-10}$

Question 7

The equation $(p-1)x^2 + 4x = 5-p$ has no real roots when

- A. $p^2 - 6p + 6 < 0$
- B. $p^2 - 6p + 1 > 0$
- C. $p^2 - 6p - 6 < 0$
- D. $p^2 - 6p + 1 < 0$
- E. $p^2 - 6p + 6 > 0$

